Implement and demonstrate the FIND-S algorithm for finding the most specific hypothesis based on a given set of training data samples.

Dataset 1: Play Tennis

Sky AirTemp Humidity Wind Water Forecast PlayTennis

Sunny Warm Normal Strong Warm Same Yes

Sunny Warm High Strong Warm Same Yes

Rainy Cold High Strong Warm Change No

Sunny Warm High Strong Cool Same Yes

Rainy Warm Normal Weak Warm Same No

Sunny Warm Normal Weak Warm Same Yes

Target Attribute: PlayTennis

CODE:

```
import pandas as pd
# Load Dataset
data = pd.read_csv("play_tennis.csv")
print("Training Data:\n")
print(data)
concepts = data.iloc[:, :-1].values
target = data.iloc[:, -1].values
# Initialize the most specific hypothesis
hypothesis = ['0'] * len(concepts[0])
# FIND-S Algorithm
for i in range(len(target)):
    if target[i] == "Yes":
        for j in range(len(hypothesis)):
            if hypothesis[j] == '0':
                hypothesis[j] = concepts[i][j]
            elif hypothesis[j] != concepts[i][j]:
                hypothesis[j] = '?'

print("\nMost Specific Hypothesis:")
```

print(hypothesis)

Output:

```
Training Data:

     Sky AirTemp Humidity    Wind Water Forecast PlayTennis
0  Sunny    Warm   Normal  Strong  Warm     Same        Yes
1  Sunny    Warm     High  Strong  Warm     Same        Yes
2  Rainy    Cold     High  Strong  Warm   Change         No
3  Sunny    Warm     High  Strong  Cool   Change        Yes
4  Rainy    Warm   Normal    Weak  Warm     Same         No
5  Sunny    Warm   Normal    Weak  Cool     Same        Yes

Most Specific Hypothesis:
['Sunny', 'Warm', '?', '?', '?', '?']
```

Dataset 2: Loan Approval

Income CreditScore Employment Property Age LoanApproved

High Good Permanent Yes Young Yes

High Good Permanent No Middle Yes

Low Poor Temporary No Young No

Medium Good Permanent Yes Middle Yes

High Average Temporary Yes Old No

High Good Permanent Yes Middle Yes

Target Attribute: LoanApproved

CODE:

```
import pandas as pd
data = pd.read_csv("loan_approval.csv")
print(data)
concepts = data.iloc[:, :-1].values
target = data.iloc[:, -1].values
hypothesis = ['0'] * len(concepts[0])
for i in range(len(target)):
    if target[i] == "Yes":
        for j in range(len(hypothesis)):
            if hypothesis[j] == '0':
                hypothesis[j] = concepts[i][j]
```

```
            elif hypothesis[j] != concepts[i][j]:
                hypothesis[j] = '?'
print("\nMost Specific Hypothesis")
print(hypothesis)
```

OUTPUT:

```
   Income CreditScore Employment Property     Age LoanApproved
0    High        High  Permanent     Good   Young          Yes
1     Low        Good  Temporary     Poor  Middle           No
2    High     Average  Permanent     Good   Young          Yes
3  Medium        High  Temporary     Good     Old           No
4    High        High  Permanent     Good  Middle          Yes

Most Specific Hypothesis
['High', '?', 'Permanent', 'Good', '?']
```

Dataset 3: Student Placement

CGPA Communication Internship Programming Aptitude Placed

High Good Yes Good High Yes

High Excellent Yes Good High Yes

Medium Average No Average Medium No

High Good Yes Excellent High Yes

Low Poor No Average Low No

High Good Yes Good Medium Yes

Target Attribute: Placed

CODE:

```
import pandas as pd
data = pd.read_csv("student_placement.csv")
print(data)
concepts = data.iloc[:, :-1].values
target = data.iloc[:, -1].values
hypothesis = ['0'] * len(concepts[0])
for i in range(len(target)):
    if target[i] == "Yes":
```

```
        for j in range(len(hypothesis)):
            if hypothesis[j] == '0':
                hypothesis[j] = concepts[i][j]
            elif hypothesis[j] != concepts[i][j]:
                hypothesis[j] = '?'
print("\nMost Specific Hypothesis")
print(hypothesis)
```

OUTPUT:

```
...      CGPA Communication Internship Programming Aptitude Placed
   0    High          Good        Yes        Good     High    Yes
   1    High     Excellent        Yes        Good     High    Yes
   2  Medium       Average         No     Average   Medium     No
   3    High          Good        Yes   Excellent     High    Yes
   4     Low          Poor         No     Average      Low     No
   5    High          Good        Yes        Good   Medium    Yes

   Most Specific Hypothesis
   ['High', '?', 'Yes', '?', '?']
```

Dataset 4: Disease Diagnosis

Fever Cough Headache BodyPain Fatigue Disease

Yes Yes Yes Yes Yes Positive

Yes Yes No Yes Yes Positive

No Yes Yes No No Negative

Yes Yes Yes No Yes Positive

No No Yes Yes No Negative

Yes Yes No No Yes Positive

Target Attribute: Disease

CODE:

```
import pandas as pd
data = pd.read_csv("disease.csv")
print(data)
concepts = data.iloc[:, :-1].values
```

```
target = data.iloc[:, -1].values
hypothesis = ['0'] * len(concepts[0])
for i in range(len(target)):
    if target[i] == "Positive":
        for j in range(len(hypothesis)):
            if hypothesis[j] == '0':
                hypothesis[j] = concepts[i][j]
            elif hypothesis[j] != concepts[i][j]:
                hypothesis[j] = '?'
print("\nMost Specific Hypothesis")
print(hypothesis)
```

OUTPUT:

```
  Fever Cough Headache BodyPain Fatigue   Disease
0   Yes   Yes      Yes      Yes     Yes  Positive
1   Yes   Yes       No      Yes     Yes  Positive
2    No   Yes      Yes       No      No  Negative
3   Yes   Yes      Yes       No     Yes  Positive
4    No    No      Yes      Yes      No  Negative
5   Yes   Yes       No       No     Yes  Positive

Most Specific Hypothesis
['Yes', 'Yes', '?', '?', 'Yes']
```

Dataset 5: Email Spam Detection

ContainsLink ContainsOffer UnknownSender Attachment ManyRecipients Spam

Yes Yes Yes No Yes Yes

Yes Yes Yes Yes Yes Yes

No No No No No No

Yes No Yes No Yes Yes

No Yes No Yes No No

Yes Yes Yes No No Yes

CODE:

```
import pandas as pd
data = pd.read_csv("spam.csv")
print(data)
concepts = data.iloc[:, :-1].values
target = data.iloc[:, -1].values
```

```
hypothesis = ['0'] * len(concepts[0])
for i in range(len(target)):
    if target[i] == "Yes":
        for j in range(len(hypothesis)):
            if hypothesis[j] == '0':
                hypothesis[j] = concepts[i][j]
            elif hypothesis[j] != concepts[i][j]:
                hypothesis[j] = '?'
print("\nMost Specific Hypothesis")
print(hypothesis)
```

OUTPUT:

```
  ContainsLink ContainsOffer UnknownSender Attachment ManyRecipients Spam
0          Yes           Yes           Yes         No            Yes  Yes
1          Yes           Yes           Yes        Yes            Yes  Yes
2           No            No            No         No             No   No
3          Yes            No           Yes         No            Yes  Yes
4           No           Yes            No        Yes             No   No
5          Yes           Yes           Yes         No             No  Yes

Most Specific Hypothesis
['Yes', '?', 'Yes', '?', '?']
```